

Andrea Gemelli

AI Research Scientist & Engineer

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andreamelli.me | [GitHub](#) | [LinkedIn](#) | [Google Scholar](#) | [Hugging Face](#)

Professional Summary

AI Research Scientist with a **PhD in Document AI** and 6+ years working where research meets production. I publish at top-tier venues (**ECCV**, **ICASSP**, **IJDAR**; **150+ citations**) and ship what I publish: training and deploying **Vision-Language Models (VLMs)** up to **32B parameters**, building the evaluation harnesses and pipelines that keep them reliable, and designing **multi-agent orchestration** systems. Open-source models and datasets have passed **20K+ downloads** on Hugging Face. Expert in **Python** and **PyTorch**.

Experience

AI Research Scientist — **Kleio**

June 2026 – Present

Paris, France

- Optimized a core orchestration component with an automatic prompt-optimization pipeline based on GEPA, raising internal benchmark pass rate by roughly **1.5×** and shipping the improvement across the full customer base.
- Built a research harness to automatically monitor, benchmark, optimize, and deploy agents and tools into production, closing the loop from measurement to deployed change.
- Design and evaluate **multi-agent architectures** for the agent-orchestration platform; partner with product and engineering to define the research roadmap and establish rigorous applied-research practice at team scale.

AI Research Scientist — **LetXbe**

Dec 2023 – April 2026

Paris, France

- Trained a proprietary **document information-extraction VLM** matching commercial-API performance while keeping all data in-house.
- Led the R&D roadmap for a 3-person research team and supervised 3 interns, resulting in published papers.
- Fine-tuned (**QLoRA**) and benchmarked open-source VLMs up to **32B parameters** on proprietary DocVQA datasets of **up to 100K documents**.
- Built an **end-to-end ML pipeline** automating preprocessing, data-quality checks, training, evaluation, and deployment; raised production precision from **85% to 95%** and drove **3×** **business-pipeline growth**.
- Open-sourced fine-tuned Qwen and Mistral models for DocVQA and the **BoundingDocs** dataset (20K+ downloads on Hugging Face).
- Ran a monthly internal reading group.

Education

PhD in Document AI — University of Florence

Florence, Italy & Barcelona, Spain | 2020 – 2023

- Awarded the **Doctor Europaeus** title (Feb 2024) for a thesis on Document AI and Graph Neural Networks.
- Published **8 peer-reviewed papers** in top-tier venues and journals (ECCV; oral at ICASSP; IJDAR and PRL).
- Visiting Researcher** at the **Computer Vision Center** (Barcelona) for one year, leading to a publication at ECCV.
- Summer schools: **Cambridge ELLIS 2022** (50% acceptance) and **ICVSS 2023** (30% acceptance).
- Supervised BSc and MSc theses on document AI and deep learning, mentoring students through to graduation.

MSc in Computer Science and Engineering — University of Florence

Florence, Italy | 2017 – 2020

- Graduated with **full marks (110/110)**. Thesis completed during an Erasmus+ Traineeship at the L3i lab, University of La Rochelle.
- Erasmus+ Study exchange at Ghent University; coursework in Machine Learning, Software Hacking, and Parallel Computing.
- Data Scientist at **Alias2k**, developing AI solutions to enhance company products.

Publications

Full list on [Google Scholar](#) (150+ citations).

- **IJDAR 2025:** [BoundingDocs: a Unified Dataset for Document Question Answering with Spatial Annotations](#).
- **ICCV 2025 (VisionDocs Workshop, Poster):** [Towards Reliable and Interpretable Document Question Answering via VLMs](#) (arXiv).
- **ICASSP 2024 (Oral):** [Structure Matters: Analyzing Videos Via Graph Neural Networks for Social Media Platform Attribution](#).
- **IJDAR 2024:** [Datasets and annotations for layout analysis of scientific articles](#).
- **PRL 2023:** [Automatic generation of scientific papers for data augmentation in document layout analysis](#).
- **DocEng 2023:** [Deep-learning for dysgraphia detection in children handwritings](#).
- **IRCDL 2023:** [CTE: A Dataset for Contextualized Table Extraction](#).
- **ECCV 2022 (TiE Workshop):** [Doc2Graph: a Task-Agnostic Document Understanding Framework Based on Graph Neural Networks](#).
- **ICPR 2022:** [Graph Neural Networks and Representation Embedding for Table Extraction in PDF Documents](#).
- **S+SSPR 2022:** [Data Augmentation on Graphs for Table Type Classification](#).

Teaching

- **Lecturer, AI for Design**, University of Florence (2026): master-level lectures introducing design students to AI, LLMs, and agents, and how to apply them within the design process.
- **Invited talks**, Computer Engineering, University of Florence: [Retrieval-Augmented Generation](#) (2024) and [AI Agents](#) (2026).
- **Teaching Assistant, Fondamenti di Informatica**, Computer Engineering, University of Florence (2023).
- **Lecturer, Fondamenti di Sistemi Informativi**, Polo Universitario Aretino (2021–2022), in collaboration with Politecnico di Milano: two academic years.

Achievements & Community

- Active reviewer for the **IJDAR** journal and the **ICDAR** conference.
- **Third place** at the {Tech: Europe} AI Hackathon (Paris, 120 applicants) with [EasyMCP](#), translating OpenAPI specs to MCP servers.
- **Third place** at the Wacom x Luleå University hackathon with a handwriting-labeling game.
- **IAPR Research Scholarship**, awarded by the IAPR through its Education Committee, 2023.
- **200+ GitHub stars** across open-source projects, including [Doc2Graph](#), a framework for Document AI.
- Author of an AI/tech blog on agents, MCP, and ML best practices, published on my [portfolio](#).

Skills & Tools

ML & Modeling: PyTorch, Transformers, TRL, PEFT, DSPy, vLLM, Ollama, Gradio, NumPy, Pandas, Scikit-Learn

Dev & Tooling: Python (Expert), Git, Docker, FastAPI, FastMCP, UV (ruff & ty), agent coding harnesses (Claude Code, OpenCode)

MLOps & Experimentation: CometML, Weights & Biases, Langfuse

Cloud & Infrastructure: Google Cloud (Vertex AI, VMs, Buckets), AWS (SageMaker, S3, Lambda), Scaleway

Languages: Italian (Native), English (C1), French (B1), Spanish (B1)